AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 16 · STUDENT LESSON

Verify Numbers and Dates

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Check the source. Recalculate. Preserve unknown intent.

Verify numeric and calendar claims with complete supplied records rather than plausible replacements.

Keep one habit: Checklist → counted percentage grids → six audits → independent learner case.

MAIN LESSON · SECTION 1

Which values can we correct, and which need a source?

> Make arithmetic visible while preserving an unknown intended date.

Your verification goals

I can verify a numeric claim using its source, denominator and units.

I can separate calendar validity from the intended event date.

I can record corrections and unresolved values without guessing.

Recall source identity and scope

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Source Quality: identify the exact source, relevant time and version before using a value. A newer-looking record may concern a different question.

Your end goal

By the end, I can verify numbers and dates without inventing the intended correction.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Multimodal **Evidence**. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 16: Verify Numbers and Dates

Precise values require direct **evidence** and recalculation.
</details>



CREATOR MISSION

Cross-check dates and recompute a percentage claim.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Give calculation and source verification different jobs

> The record supplies counts and a calendar fact, but no intended meeting date.

Understand numerator, denominator and validity

A precise-looking number can still use the wrong source, denominator or unit. A denominator is the total used in a fraction: 15 passes out of 20 trials means 15/20, not 15/15. Verify the source/run ID and counted population before calculating. A percentage is the fraction multiplied by 100. Calendar validity is a different check: an impossible date can be rejected without knowing what date its author intended.

Read the complete main record

Complete fictional main record: 15 of 20 model trials passed. The draft says 70% passed and announces a meeting on June 31. Calendar fact: June has 30 days. No authoritative meeting notice is supplied. Recompute the percentage from the record; identify the invalid date and request the intended meeting date. These invented counts do not establish that any real model was tested.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use verify numbers and dates to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Citation evaluation checks source existence, exact passage support and coverage of answer claims. A real citation can still be irrelevant.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Check the meaning before pressing calculate

> Correct arithmetic on the wrong source still answers the wrong question.

The verification checklist

Check	Question to ask
Source and scope	Which record, run, population and time does the claim concern?
Numerator and denominator	What counts as a pass, and what total is used?
Units	Are quantities in compatible units before combining them?
Recalculation and rounding	Can I reproduce the exact fraction and stated rounding rule?
Calendar validity	Does this date exist in the stated calendar and year?
Intended date	Which responsible source actually states the event date?

Explain the deciding check

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Review · Earlier citation denominators

For this classroom rubric count a claim as supported only when its cited passage entails it. Separately count correct citation links. Use the number of factual claims as the support denominator.

Before applying the procedure, write its **unit**s, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

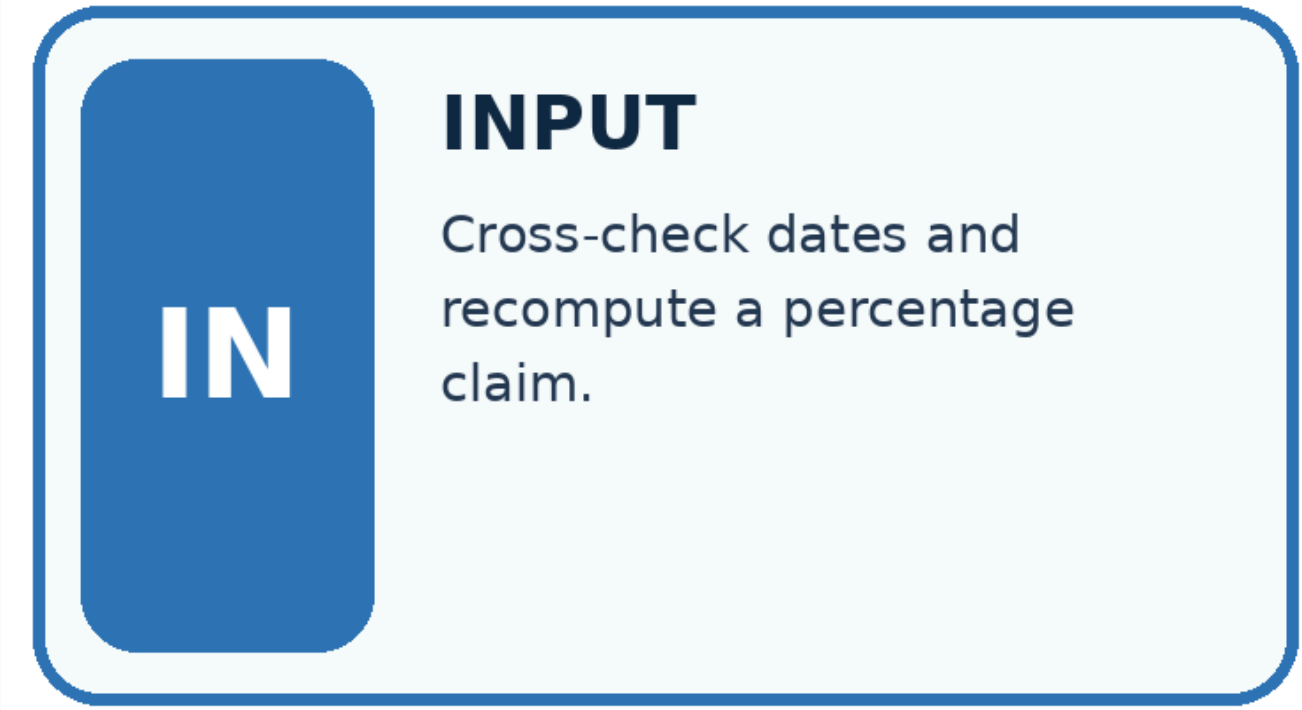
If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

▼ Earlier wording, alternatives and picture transcripts

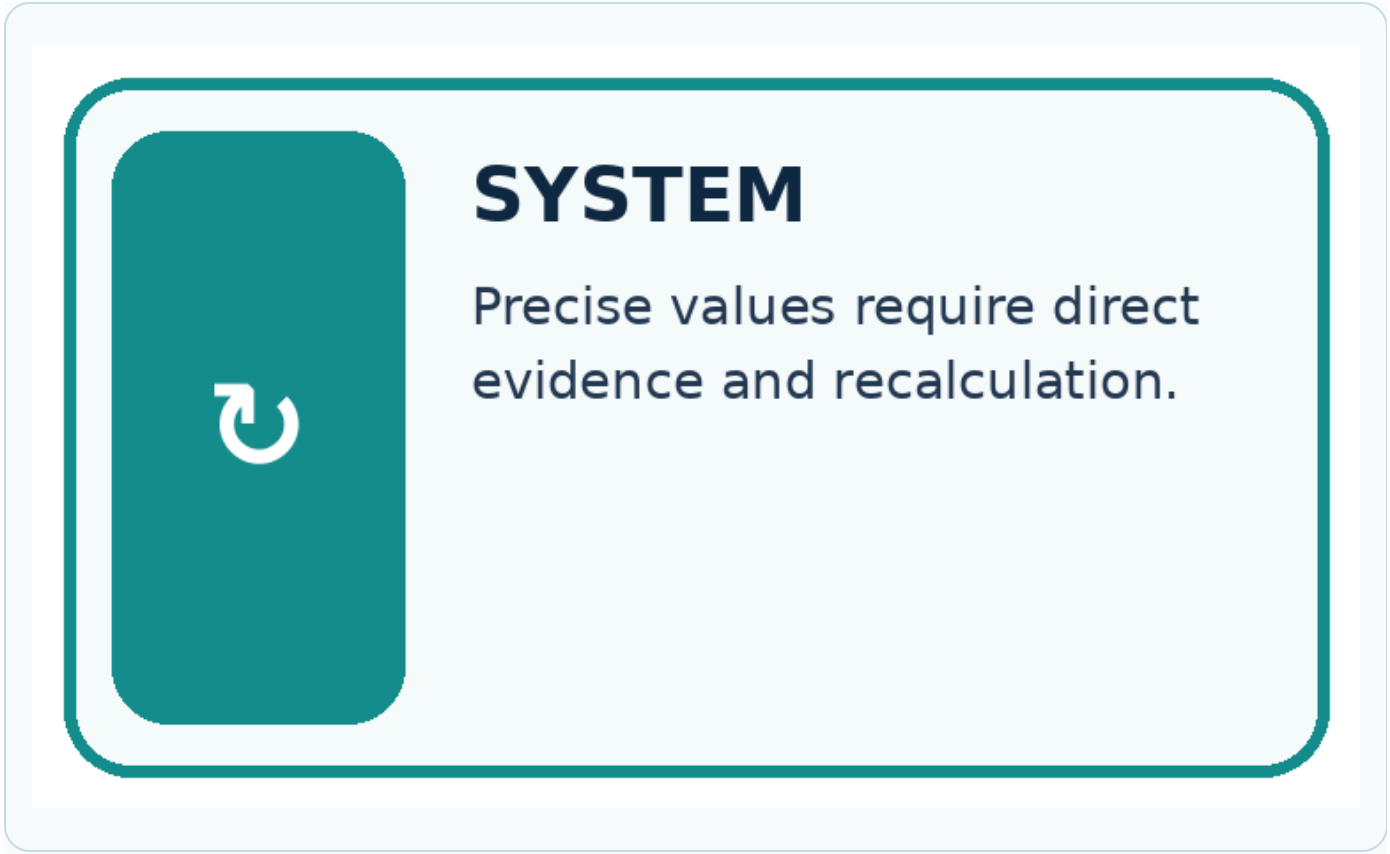
VISUAL **MECHANISM**

Input → Process → Output

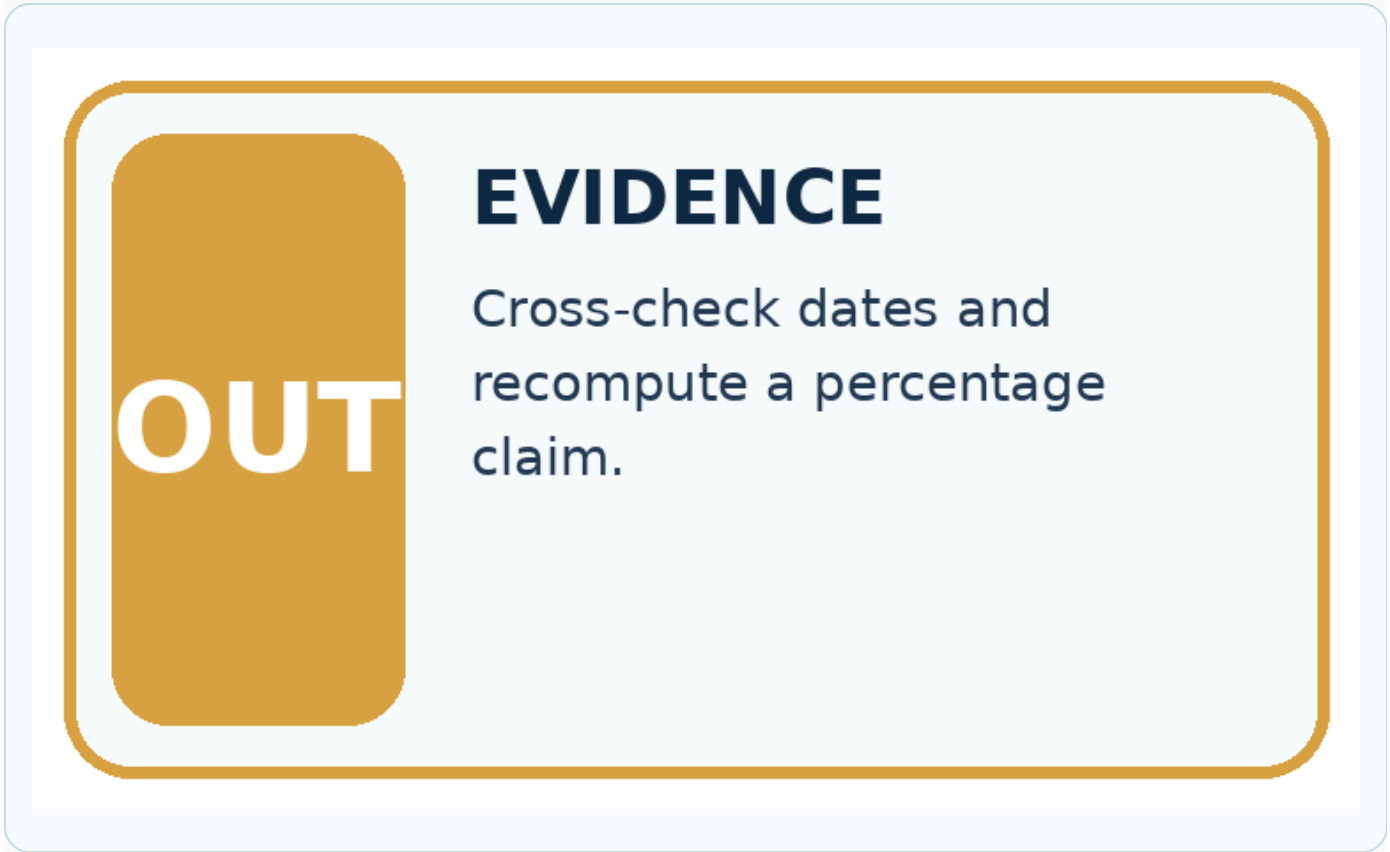
Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Precise values require direct evidence and recalculation.

↓


OUT
EVIDENCE
Cross-check dates and recompute a percentage claim.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Claim support and citation evaluation — evidence trace

INPUT AND SCOPE

An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist.

CHECK OR CALCULATION

For this classroom rubric count a claim as supported only when its cited passage entails it.

CONCLUSION AND LIMIT

Claim support is 2/4=50%; citation presence is 3/4=75%; link existence among three citations is 3/3. These measure different failures.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

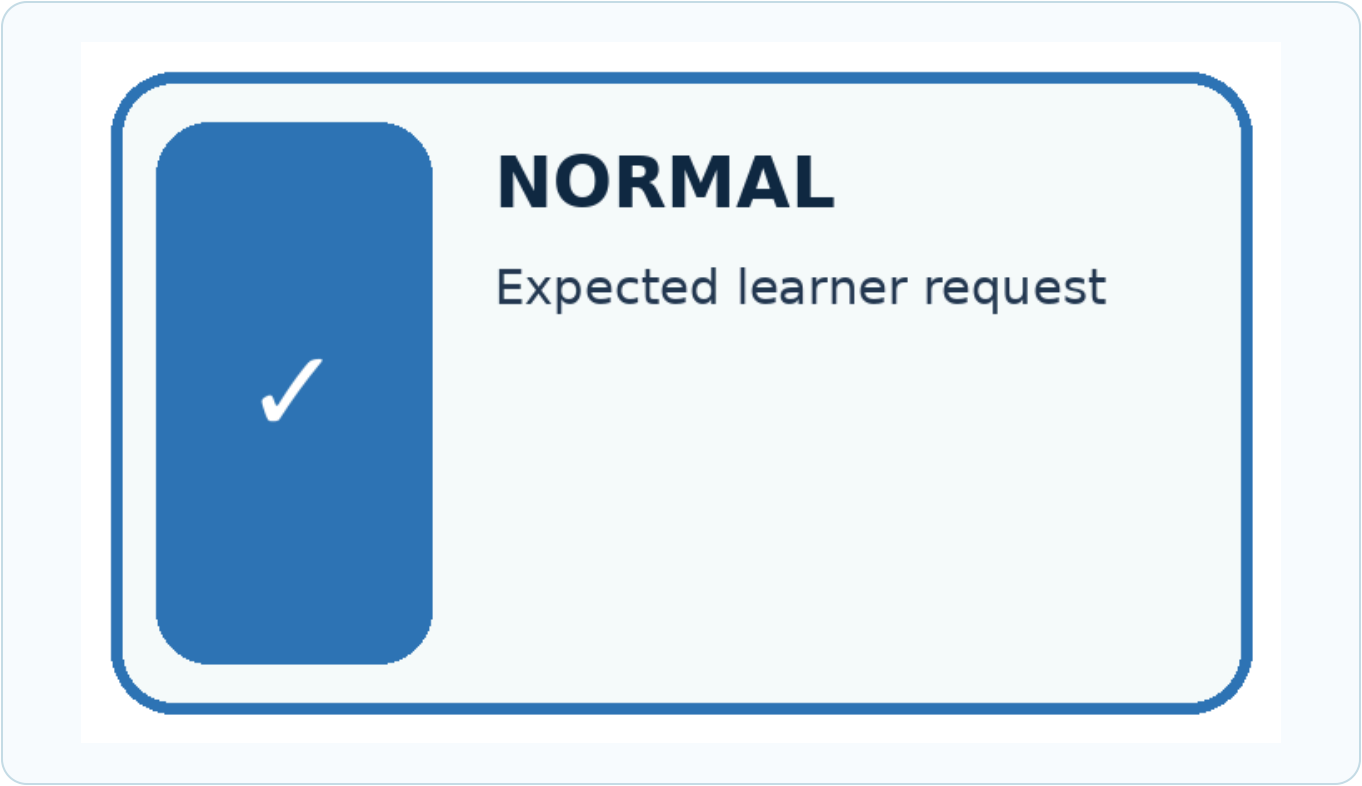
Connect the picture and the explanation

MAIN LESSON · SECTION 4

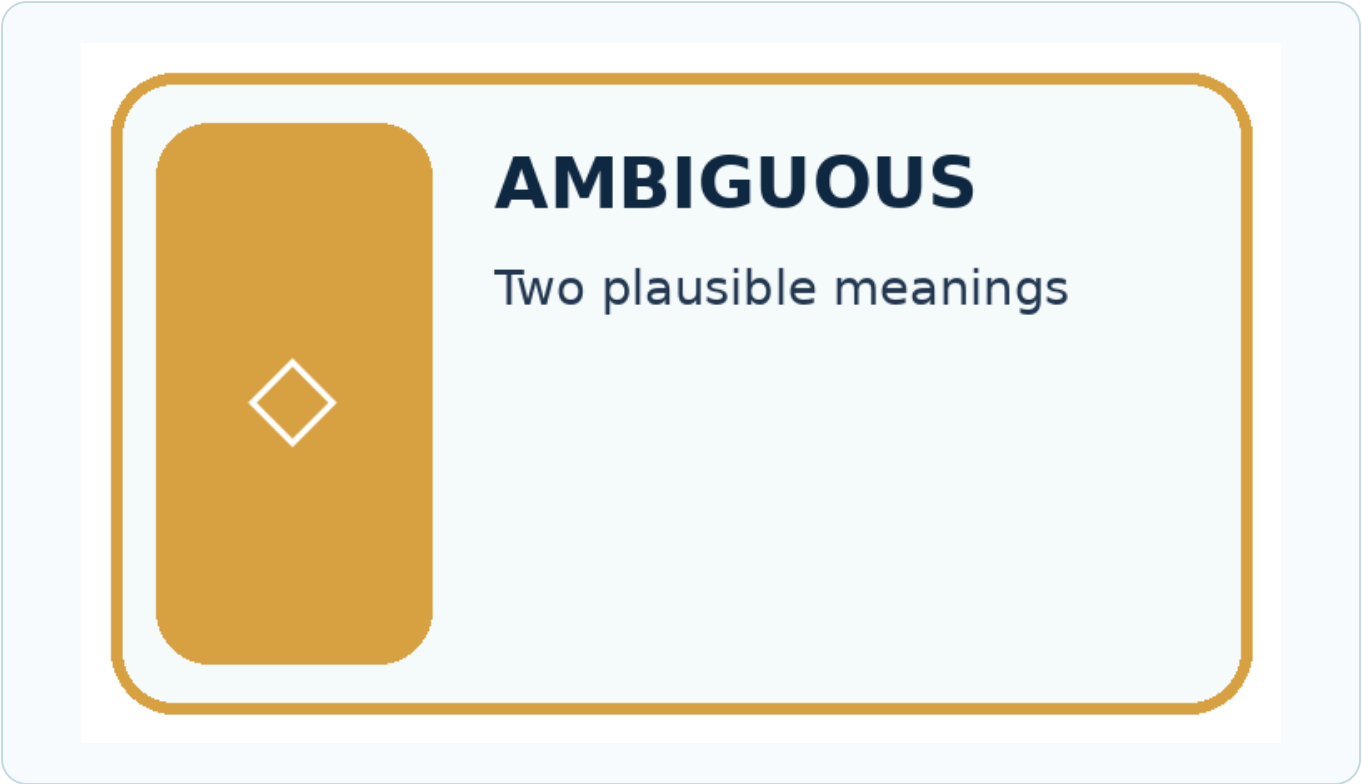
Compare correct arithmetic with supported claims

> A calendar can reject an impossible date without identifying the intended replacement.

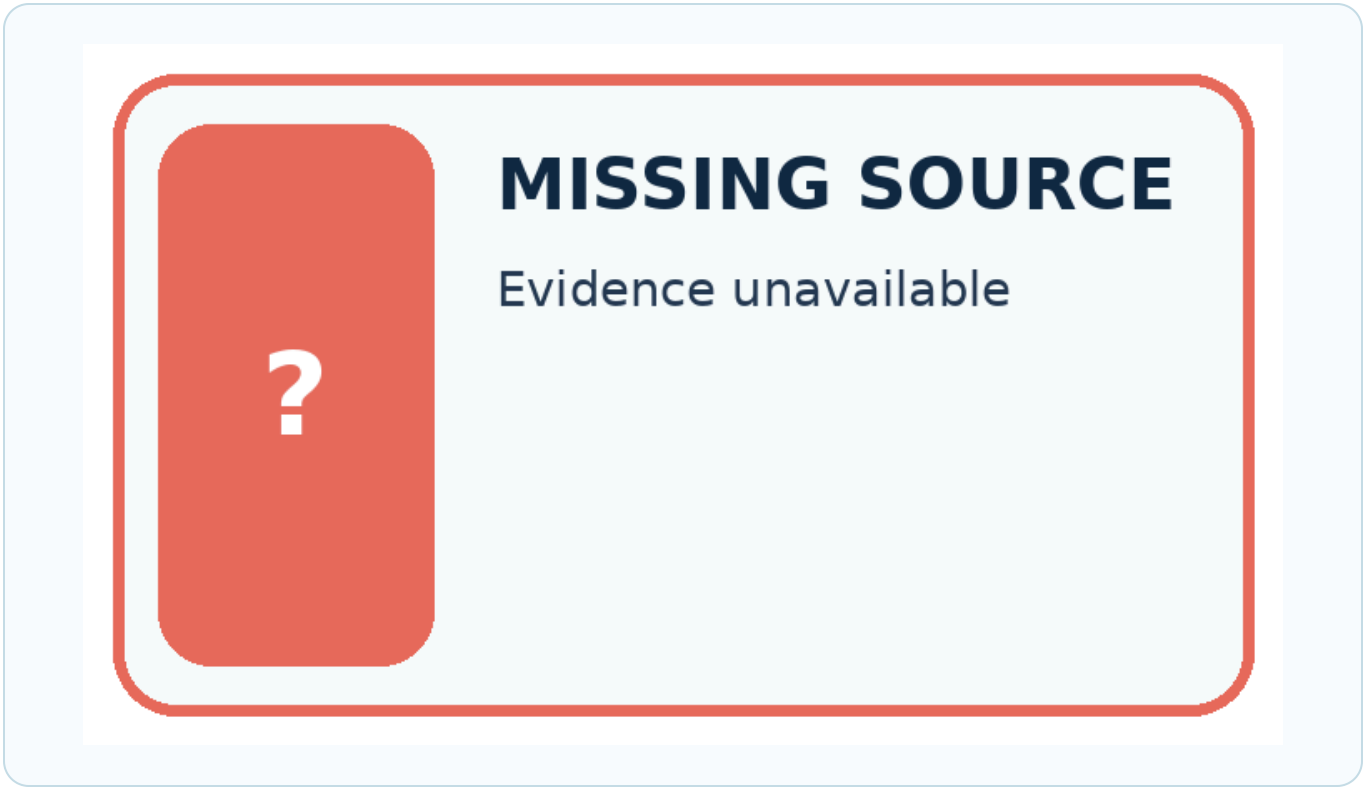
Four pictures with a purpose



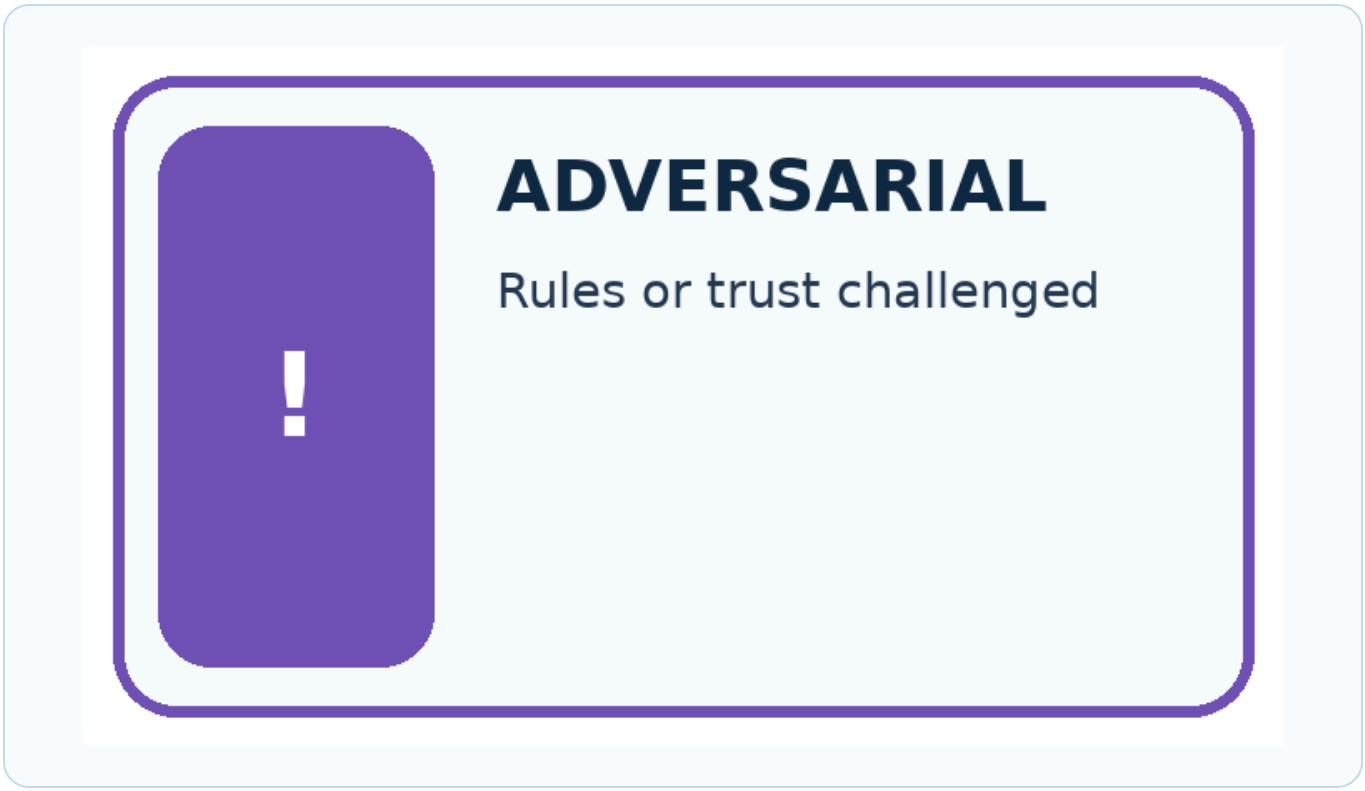
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: The count is revised to 14 passes out of 20, but no date record is added.

▼ Check after predicting

Prediction: **70%** now matches the count; June 31 remains invalid and unresolved.

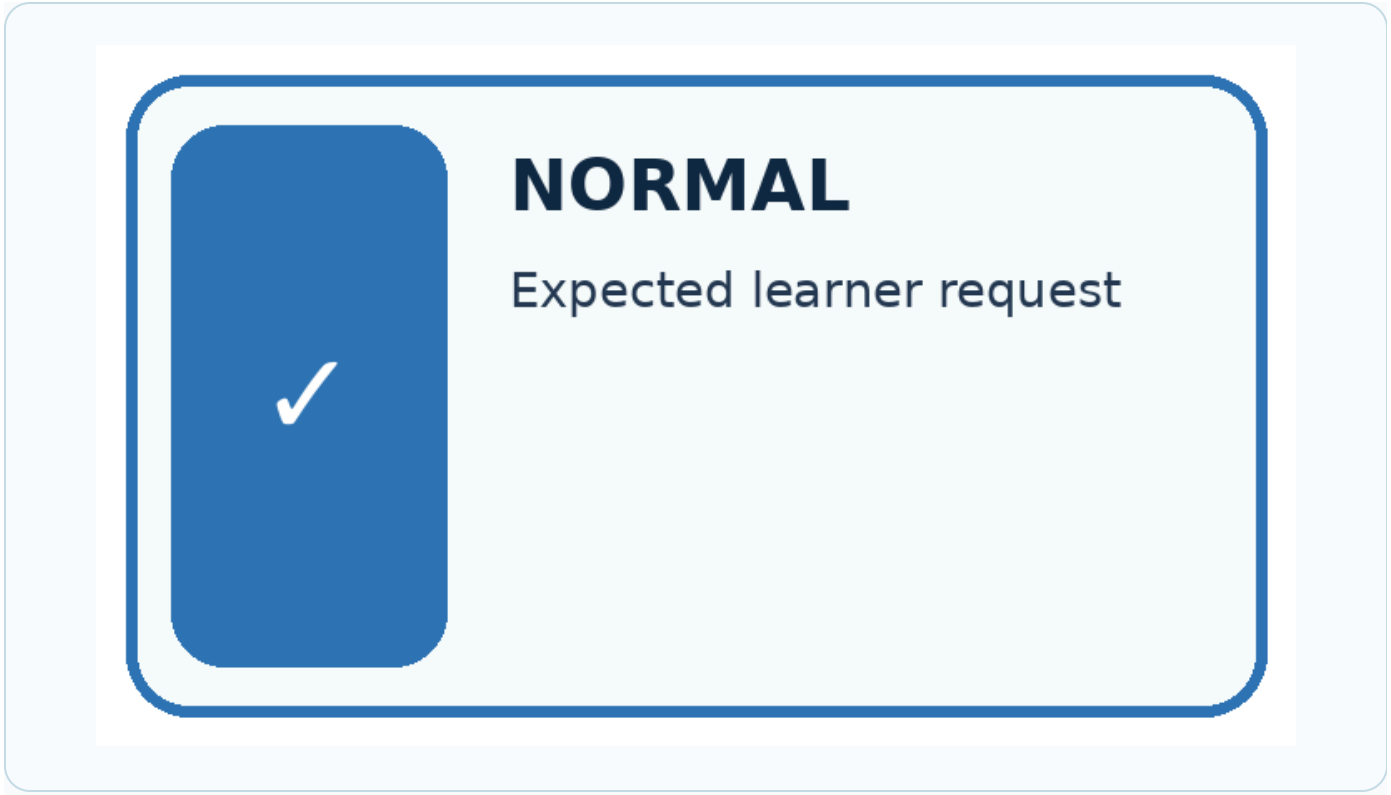
Reason: 14÷20×100=**70%**; changing a count provides no new calendar **evidence**.

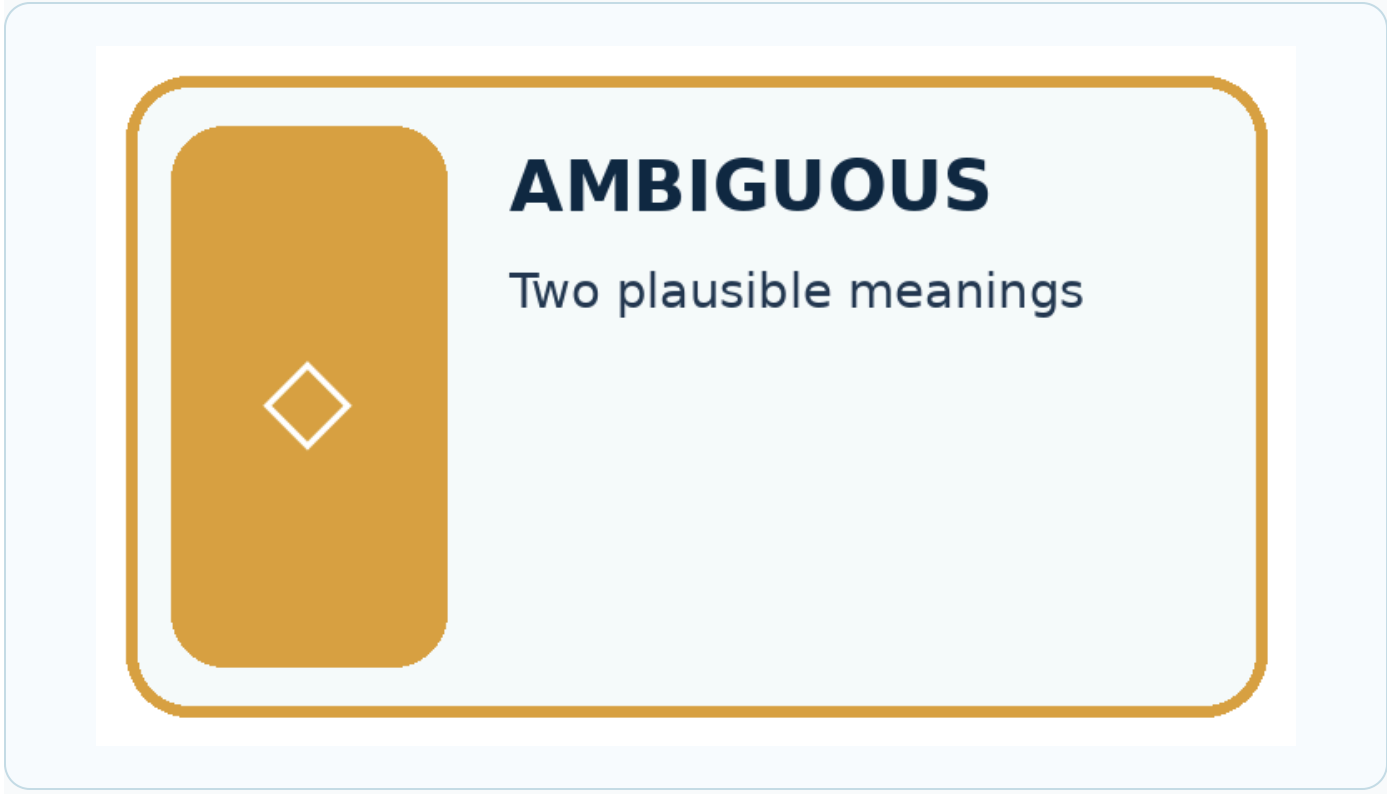
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the fraction and the separate date boundary

> Changing the count supplies no new meeting-date evidence.

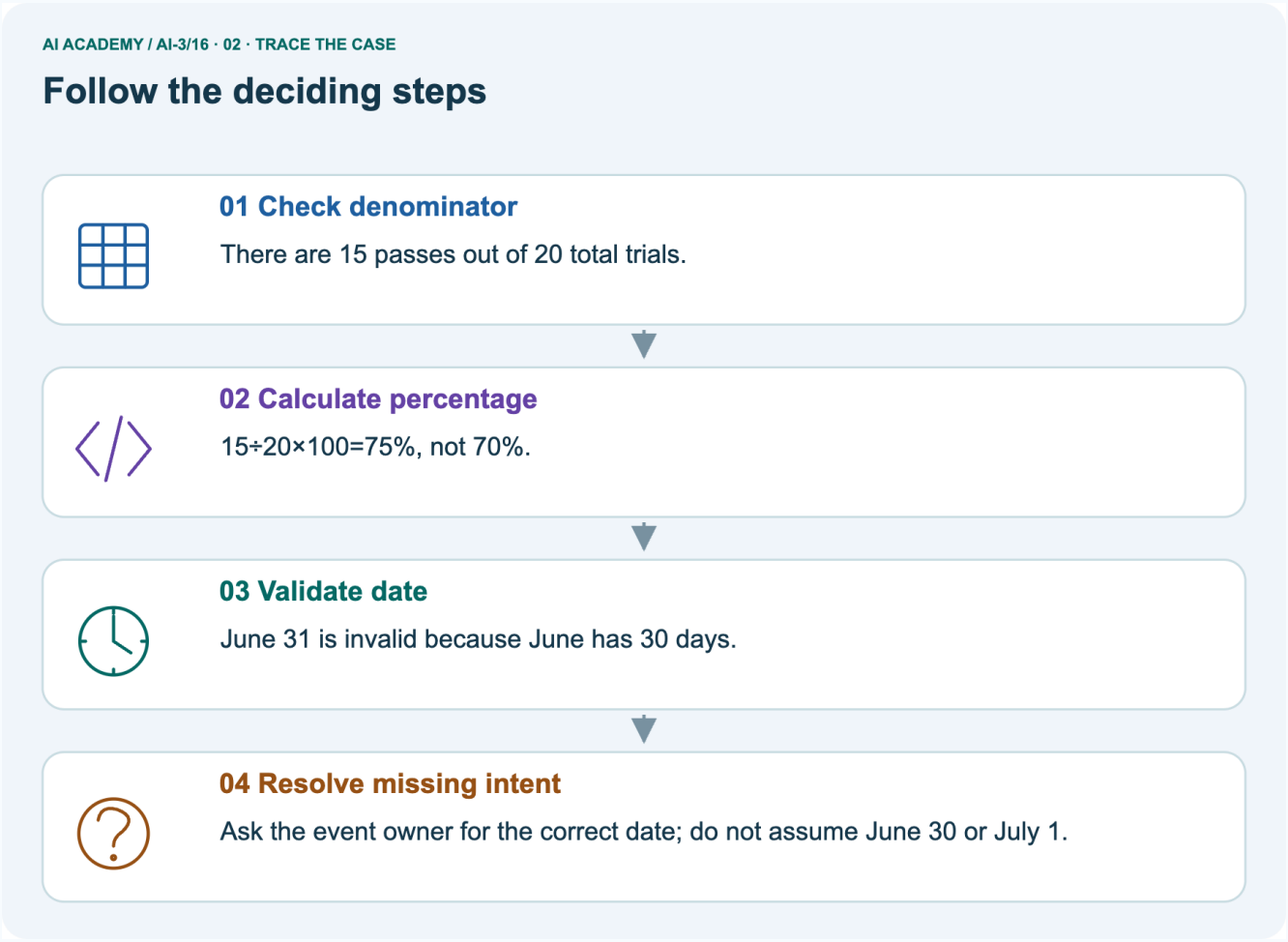
Read the complete fictional record

A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts.

Record: 15 of 20 fictional model trials passed. Draft: “**70%** passed; the meeting is on June 31.”

June has 30 days. No authoritative record gives the intended meeting date.

Trace the arithmetic and invalid date



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

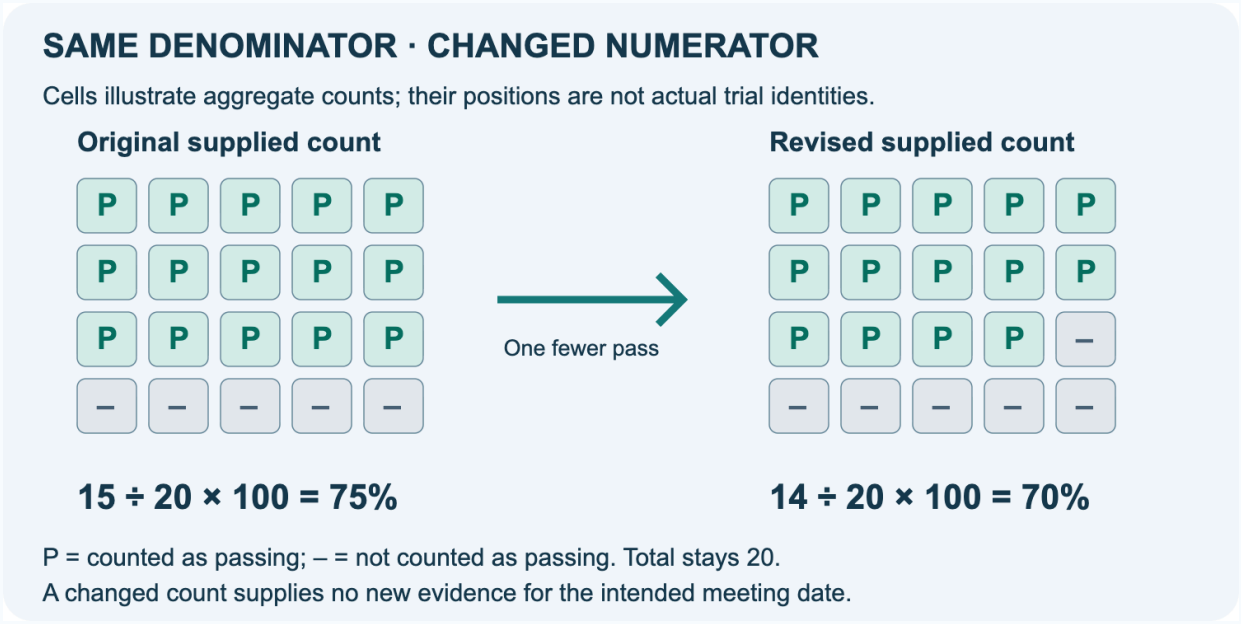
STEP	WHAT HAPPENS AND WHY
1. Check denominator	There are 15 passes out of 20 total trials.
2. Calculate percentage	15÷20×100=75%, not 70%.
3. Validate date	June 31 is invalid because June has 30 days.
4. Resolve missing intent	Ask the event owner for the correct date; do not assume June 30 or July 1.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A percentage requires the correct numerator and denominator. Calendar validation identifies an impossible date, but it cannot discover what date the author intended.

These checks use different **evidence**: arithmetic repairs the percentage from the supplied counts, while the date needs another authoritative source before publication.

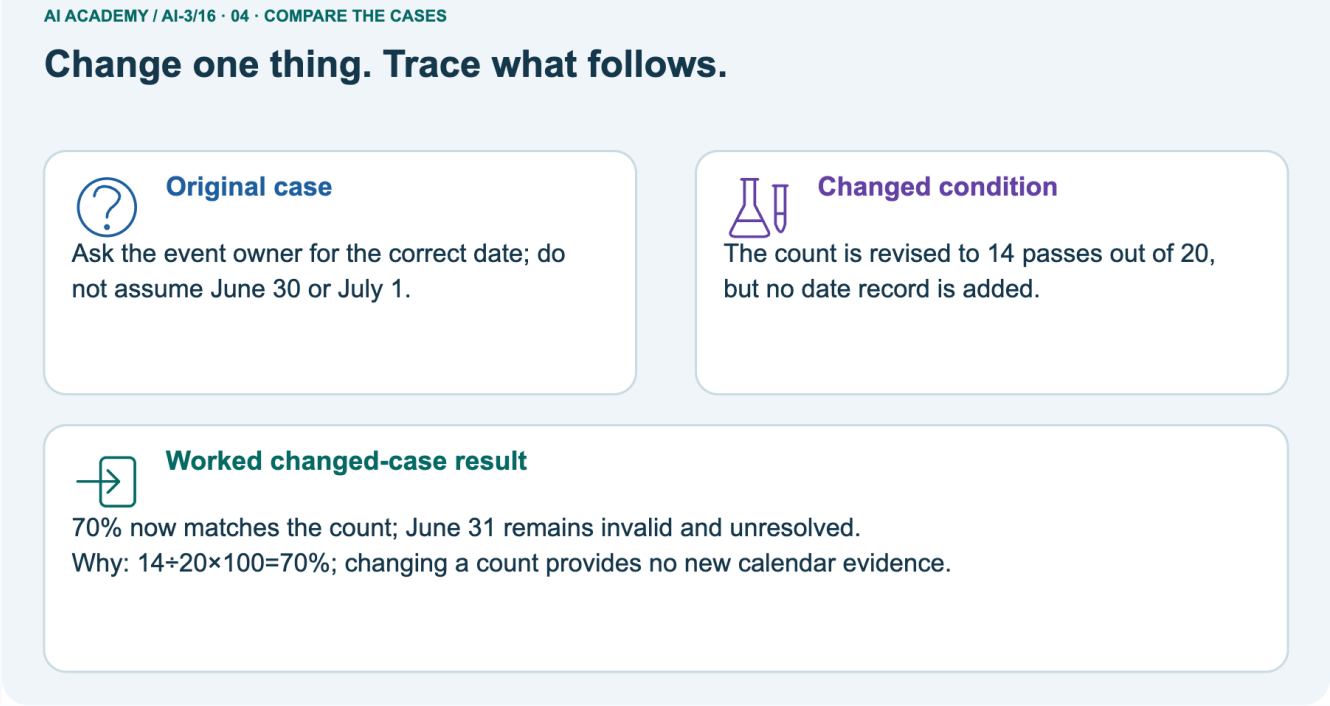
Count the percentage grids



Predict the revised count

- Change one thing: The count is revised to 14 passes out of 20, but no date record is added. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Name the source reliability limit

- This check confirms internal arithmetic and date validity, not whether the fictional trial measurements themselves were reliable.
- Read the labelled picture: Which part can be corrected immediately from existing **evidence**, and which needs follow-up?

▼ Review · Separate numeric claim

1. Claim: 18 of 24 tests passed, reported as **80%**.
2. **Recalculate**: 18 ÷ 24 = 0.75 = 75%.
3. Check denominator, **unit**, rounding, and source date.
4. Correct the claim and preserve the calculation trail.

▼ Review · Earlier citation case

An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist.

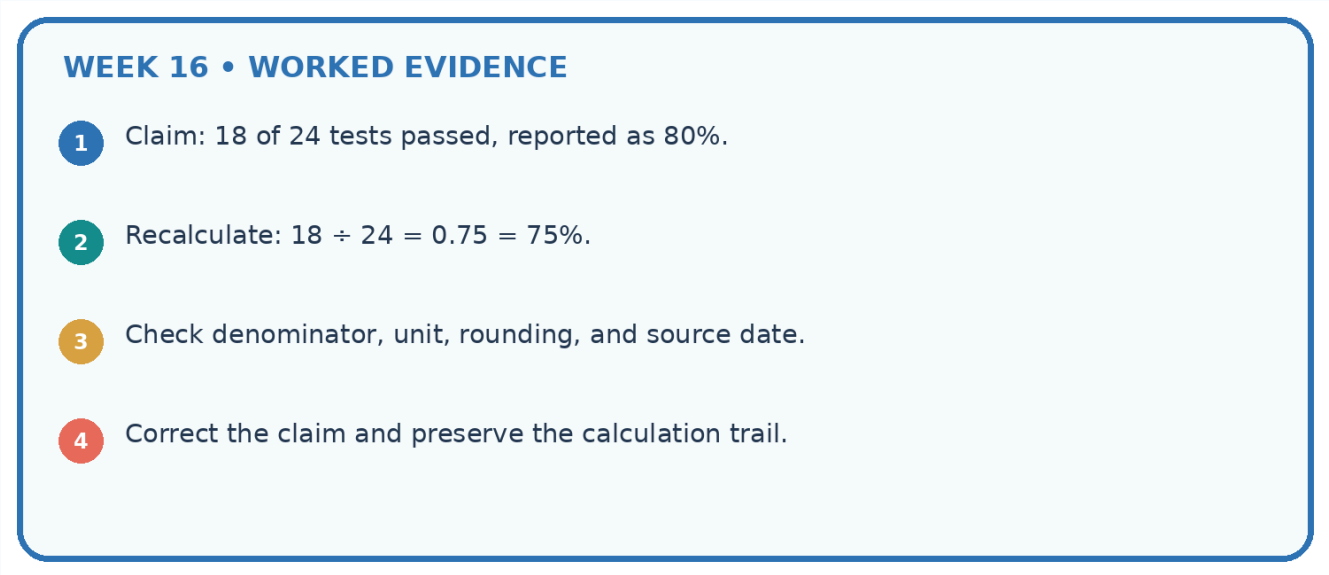
Claim support is 2/4=50%; citation presence is 3/4=75%; link existence among three citations is 3/3. These measure different failures.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

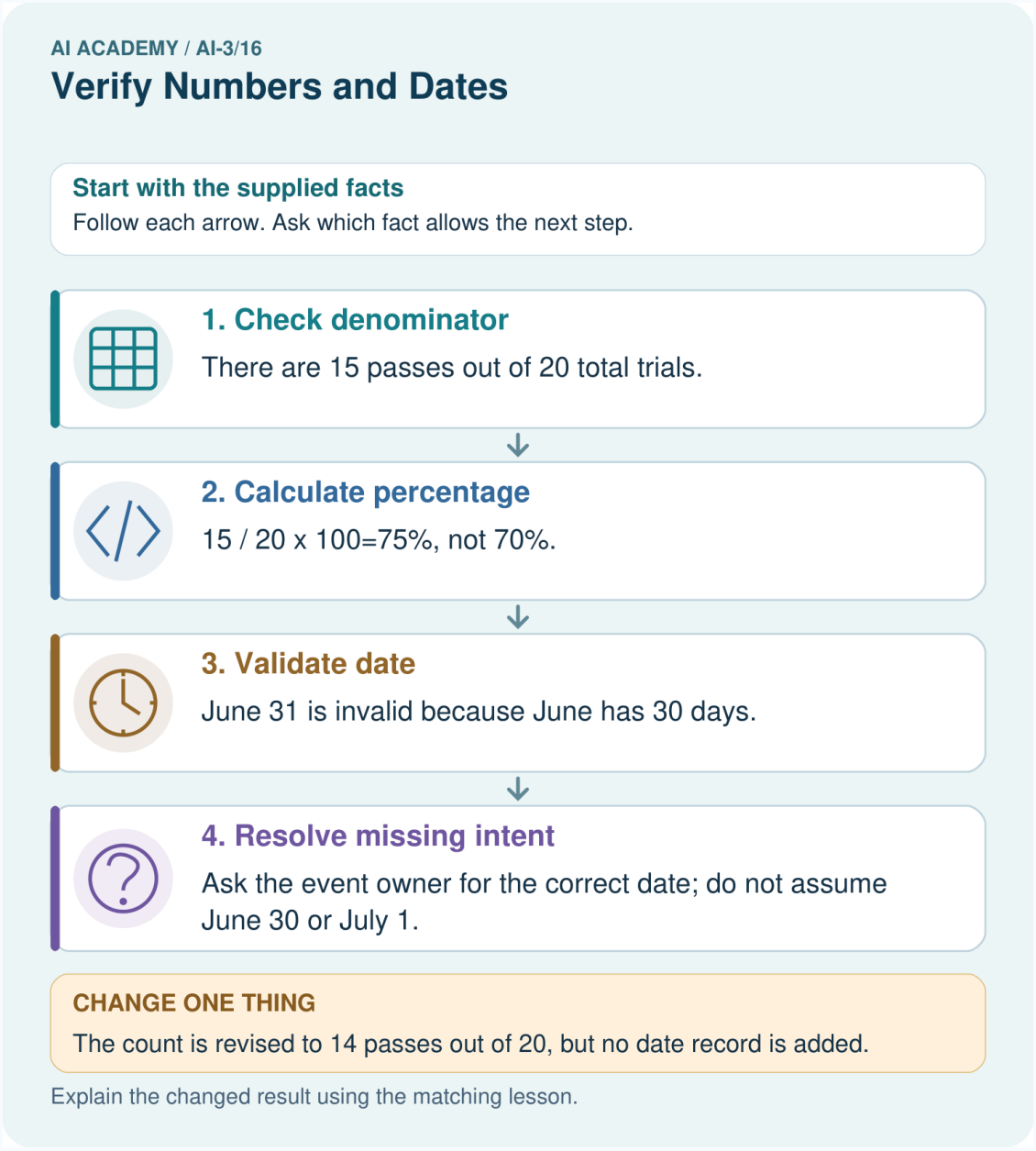
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Verify Numbers and Dates

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the unsupported replacement

> A detected error and a justified correction require different evidence.

Repair the calendar shortcut

Claim to repair: June 31 is impossible, so the meeting must be June 30. The calendar disproves June 31; it does not select June 30 rather than July 1 or another date. Ask the responsible organiser or obtain the approved notice. Similarly, correcting 15/20 to 75% does not prove that the source counts were measured reliably.

Design a revealing follow-up

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Specific numbers are reliable because they look precise. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the checks with less help

> Keep source facts, calculations and unresolved fields distinct.

Current practice and response

- Use the fictional record of 15 passes out of 20 trials and the draft 70%, meeting June 31. Show the fraction and percentage, state the supplied calendar fact, and separate the computable correction from the missing intended date. Then change only the count to 14 passes out of 20 and explain what changes and what remains unresolved.

My numeric and calendar checks

Explain the pictured boundary

- Which part can be corrected immediately from existing **evidence**, and which needs follow-up?

Explain the picture

My explanation using a pictured step and the given facts

Keep the independent task separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Review · Earlier citation practice

Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Claim support and citation evaluation”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Check your reasoning

Use the worked case for Claim support and citation evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

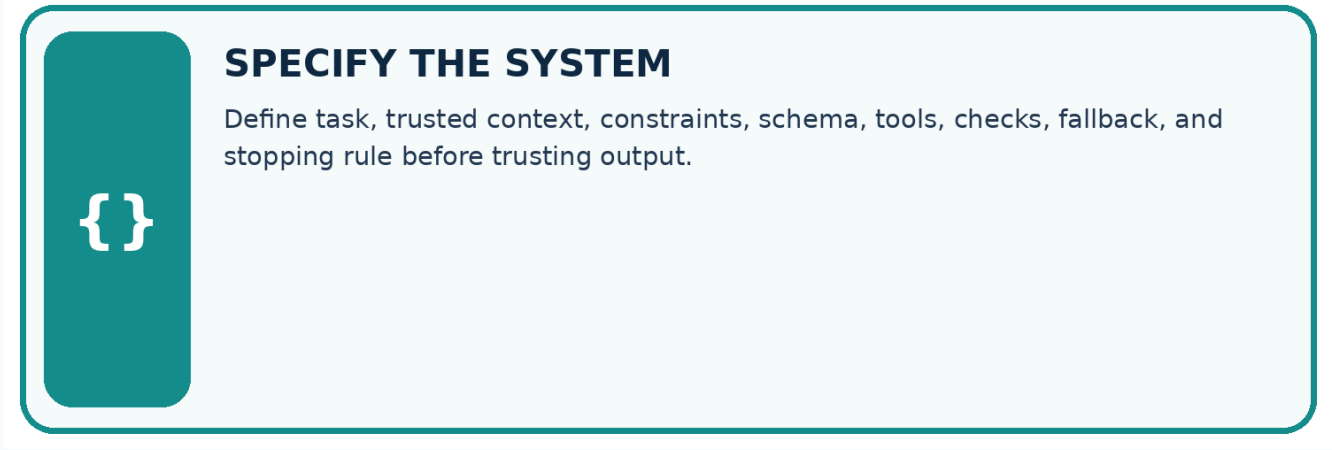
Answer on your own before discussion. If you are unsure, say which step you need help with.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Audit six complete number and date cards

> Different checks reveal different failures.

All supplied records and claims

Card	Complete fictional record and draft	Audit question
A	Record: 15/20 trials passed. Draft: 70%.	Recompute using the recorded denominator.
B	Run A: 7/10 passed. Run B: 8/10 passed. Draft cites A and claims 80%.	Check run identity before calculation.
C	Route legs: 2 km and 500 m. Draft: total 502 km. Given 1 km = 1000 m.	Convert to the same unit.
D	Draft meeting date: June 31. June has 30 days. No organiser notice supplied.	Reject invalidity without guessing intent.
E	Approved fictional notice states 2026-06-30; draft repeats 2026-06-30.	Check calendar validity and source support separately.

Card	Complete fictional record and draft	Audit question
F	Draft date: 03/04/2026. No date-format convention or original notice supplied.	Preserve ambiguity rather than assuming month/day order.

Build a correction and uncertainty record

1. For each card, record the source ID, given values and units, deciding check, supported correction and remaining uncertainty. A: 75%. B: A gives 70%; B's 80% cannot support a claim citing A. C: 500 m = 0.5 km, so total 2.5 km. D: invalid date; intended date unknown. E: the date is valid and matches the supplied approved notice; this is not external authentication.

   F: both March 4 and April 3 fit common conventions; request the original source or an explicit format. These are supplied classroom records, not real experiments. A valid date alone would not establish an event's intended date. Keep the selected independent assessment withheld.

Keep the evidence

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Audit the six supplied numeric and date cards. Check source identity, denominators, **unit** conversion, calendar validity and ambiguous formats. Record supported corrections and unresolved intent separately; these are fictional records, not live model results.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit a fresh percentage and deadline

> Keep this first attempt separate from supported practice.

Independent record and scope

A draft claims18 of24 learners is**60%**, and gives an April31 deadline. Check both statements without inventing a deadline.

Use only the supplied learner counts, claimed percentage and deadline. Show your checks and identify any additional source needed. No actual attendance data or approved deadline notice is supplied.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 16 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A report says 12 of 40 responses is 40%. **Recalculate** and state the correction with the denominator.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Route arithmetic to a calculator and dates to direct sources before presenting them.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A report says 21/28=**70%** and gives February 30 as the deadline. Correct what can be computed and identify what must be checked externally.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's correction trail

> Make each calculated value reproducible.

This week's contribution

For the Verified Study Buddy, store the source/run ID, original claim, counts and units, calculation, rounding policy and unresolved fields. For a date, store both its calendar-validity result and the source that authorises the event date. A calculator verifies an operation, not the truth of its inputs. Ask for missing evidence instead of filling a deadline with the nearest valid day.

Make Study Buddy check quantities, **unit**s, dates and source support separately; record unresolved fields instead of guessing.

Retain the checkpoint and human choice

THIS WEEK'S CHECKPOINT Route arithmetic to a calculator and dates to direct sources before presenting them.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

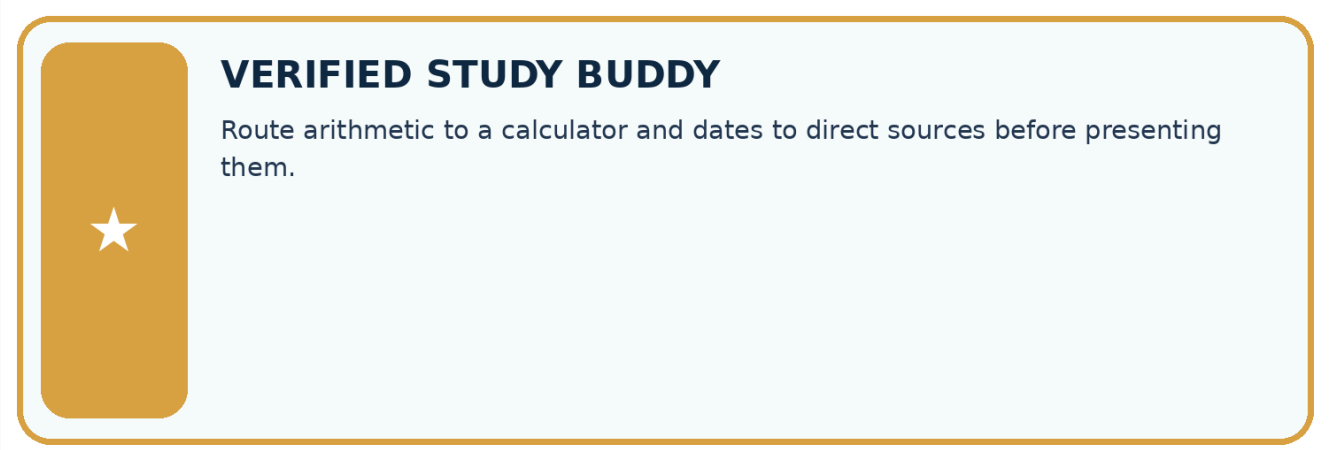
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: calculate, validate, then request missing evidence

> Preserve the source and the unresolved value.

Explain without notes

- Close your notes. Explain why a correct calculation, a valid calendar date and an authorised event date are three different checks.

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
primary record	An original record directly relevant to a claim, rather than a later summary of it.	
recalculate	Compute a quantity again from the original values or corrected inputs.	
unit	An agreed amount used to express a measurement, such as centimetres or seconds.	
as-of date	The date up to which a claim or record is stated to be current.	

Terms for the connected explanation

Word	Meaning
Entailment	Whether the cited passage supports the specific claim
Citation coverage	Which claims include source references

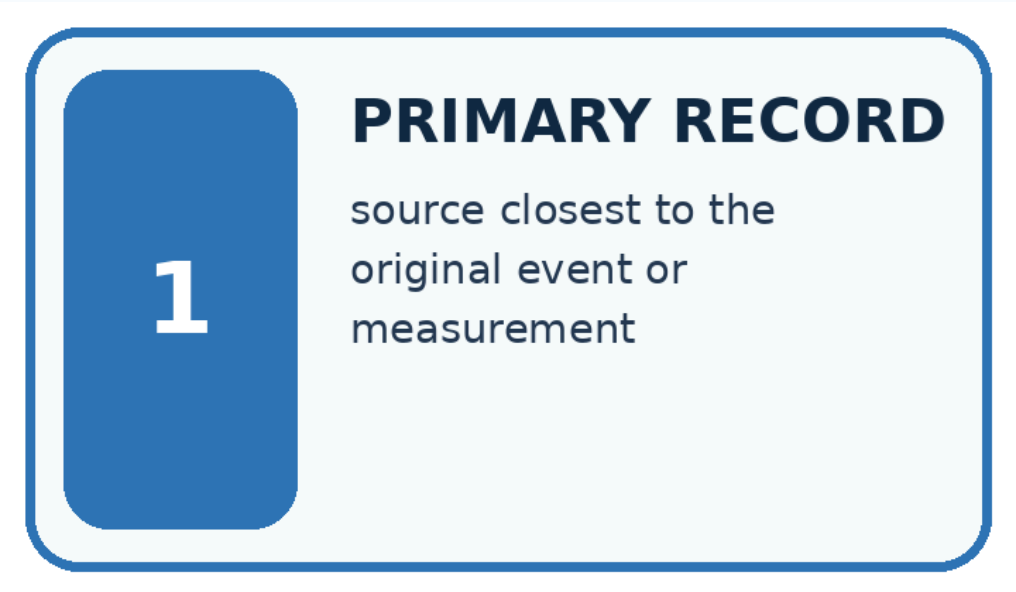
Word	Meaning
Link validity	Whether a source reference resolves to the intended record

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

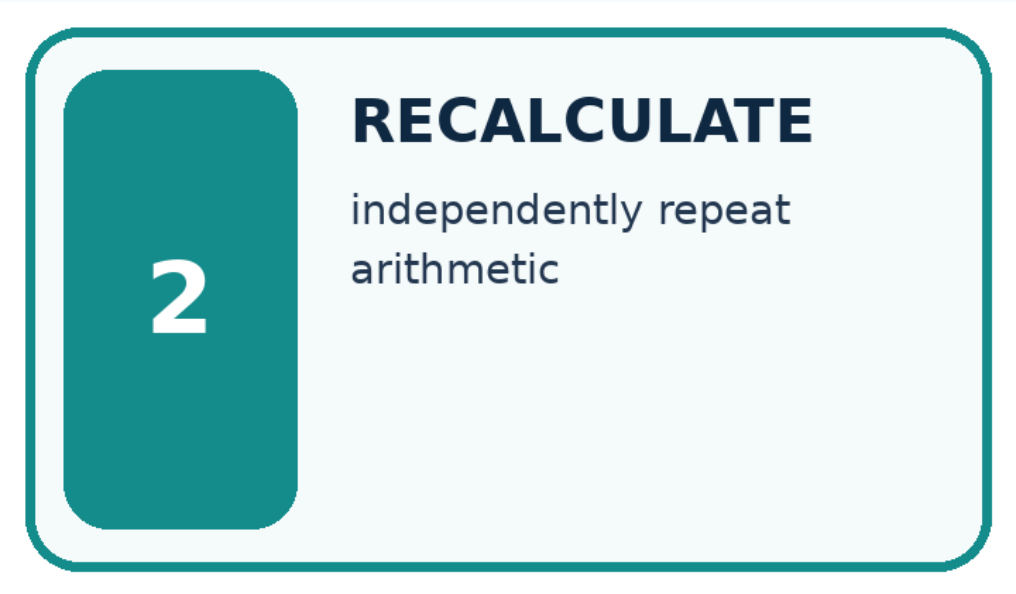
VOCABULARY LAB

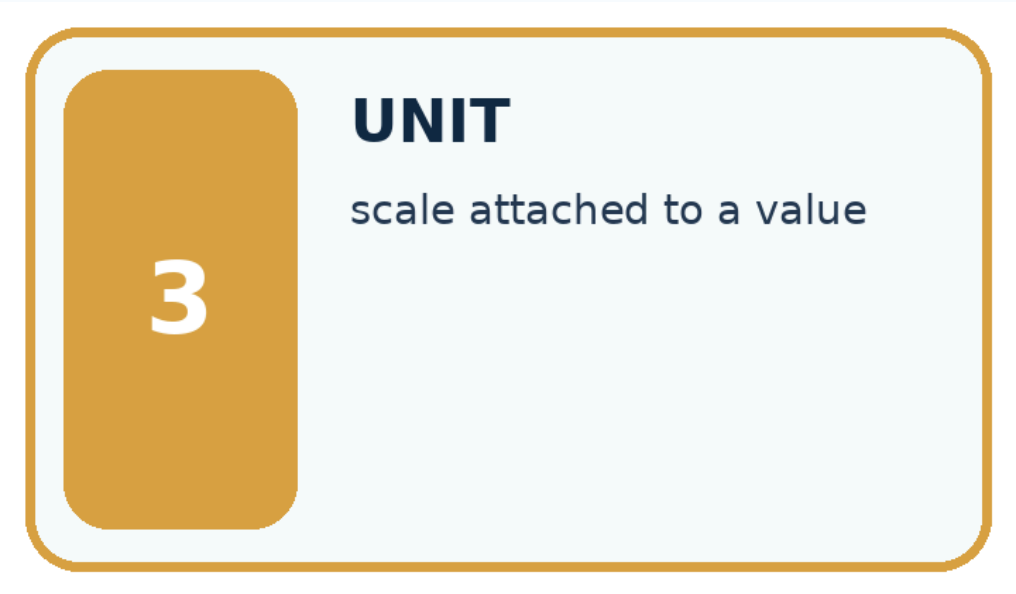
Words You Can Use to Reason

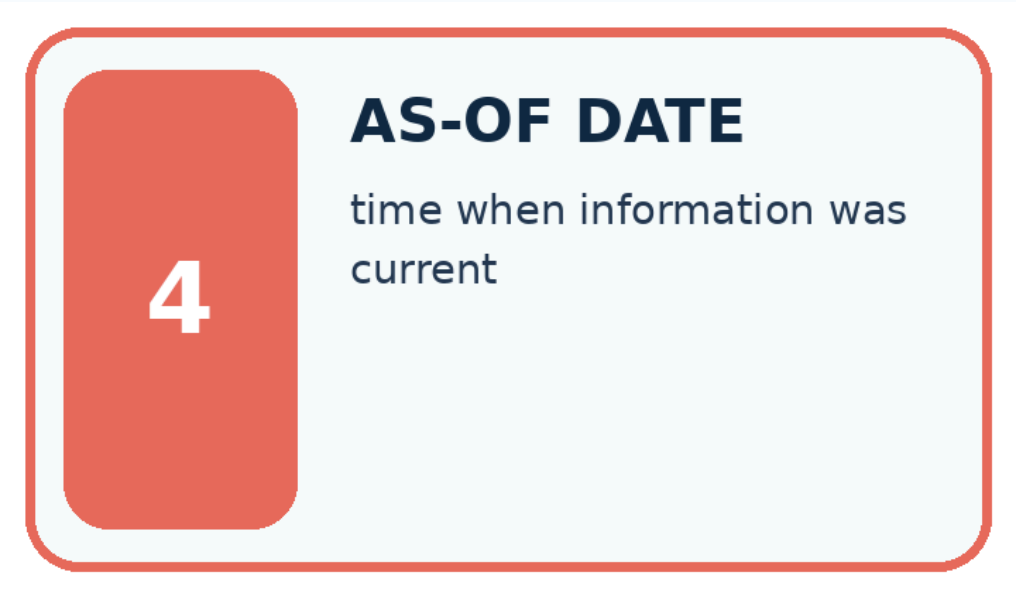
A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



</details>


2
RECALCULATE
independently repeat arithmetic


3
UNIT
scale attached to a value



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

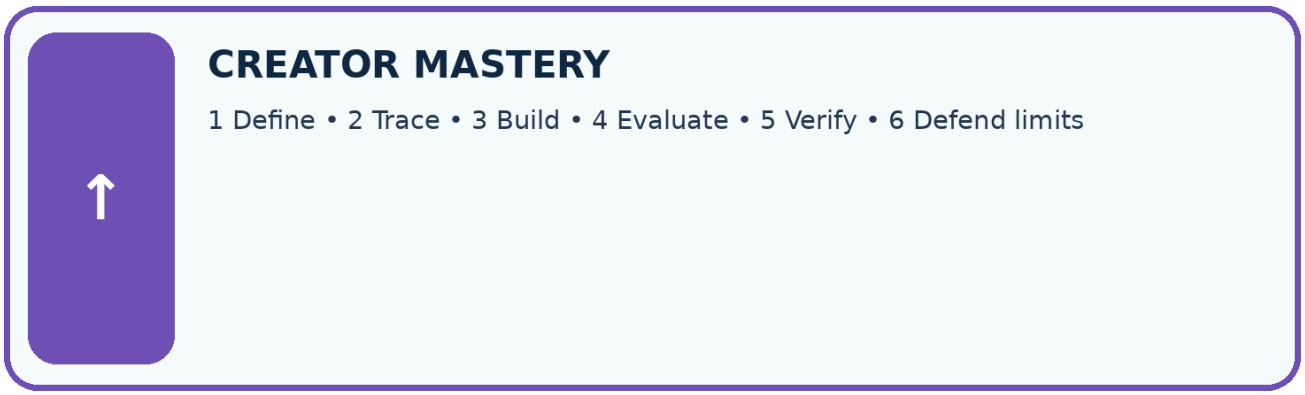
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Audit a fresh learner percentage and deadline independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

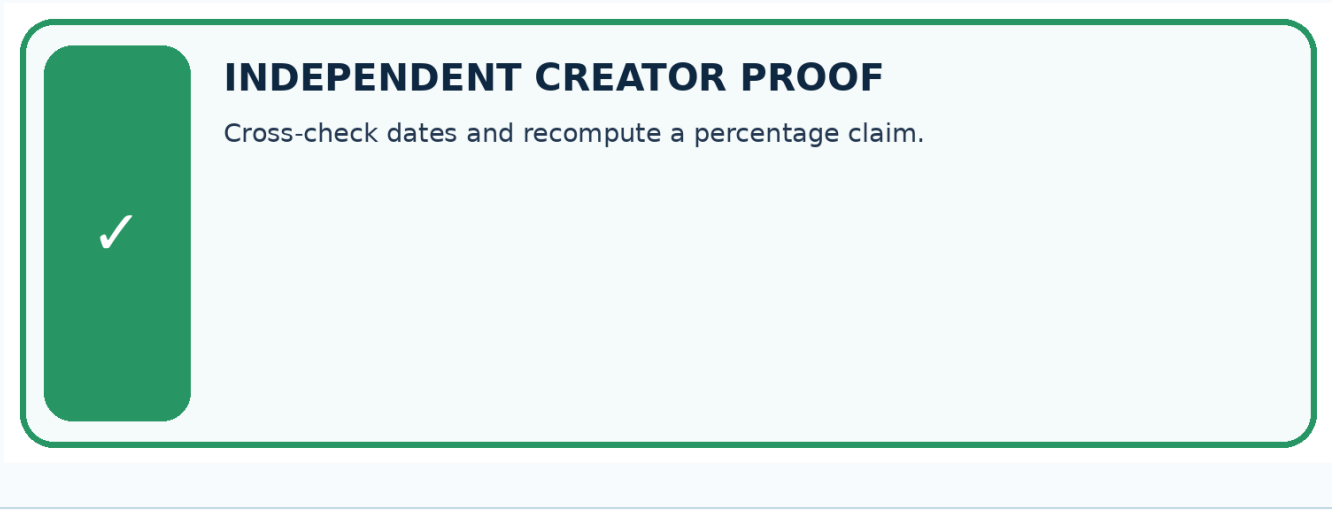
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Audit a fresh learner percentage and deadline independently is the selected independent case.



EXPLANATION CHECK Require the correct denominator, visible arithmetic, calendar check and an explicit unresolved intended date.

Write or draw your reasoning and evidence.

▼ **Extra practice: Verify a percentage independently**

REFERENCE · SECTION 13

Extra practice: Verify a percentage independently

Try the question before opening its worked explanation.

Read the supplied case and question

An answer says 15/20=**80%** and sets a deadline of February 30.

Verify both.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

15/20=75%. February 30 is invalid in the Gregorian calendar. Check the intended deadline from its authority rather than guessing.

Evidence or stage	Value or result
Correct portion	15/20
Remaining portion	5/20

▼ Earlier wording, alternatives and picture transcripts

Week 16 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Verify a percentage independently

An answer says 15/20=**80%** and sets a deadline of February 30.

Verify both.

15/20=75%. February 30 is invalid in the Gregorian calendar. Check the intended deadline from its authority rather than guessing.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of claim support and citation evaluation	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts.
- Which labelled step matches this detail? Ask the event owner for the correct date; do not assume June 30 or July 1.
- 1. Validate date
- 2. Check denominator
- 3. Resolve missing intent

- 4. Calculate percentage

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ___
>
> ___
>
> ___
>
> ___

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Resolve missing intent Ask the event owner for the correct date; do not assume June 30 or July 1. A percentage requires the correct numerator and denominator.

Calendar validation identifies an impossible date, but it cannot discover what date the author intended.

These checks use different **evidence**: arithmetic repairs the percentage from the supplied counts, while the date needs another authoritative source before publication.

Changed-case question

- Changed-case practice: The count is revised to 14 passes out of 20, but no date record is added. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. **70%** now matches the count; June 31 remains invalid and unresolved.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: **70%** now matches the count; June 31 remains invalid and unresolved. 14÷20×100=**70%**; changing a count provides no new calendar **evidence**.

Repair and save your work

Review Source Quality (ai-3/15). Goal: Explain the **mechanism** and terminology of conflicting and time-sensitive sources. Short practice: Which source fields must be checked before answering day 9?.

Return to this week and explain what you changed.

Review Multimodal **Evidence** (ai-3/13). Goal: Keep visible **evidence**, hidden contents and caption claims separate. Short practice: Why were both “exactly three” and “exactly five” unsupported before opening the box?. Return to this week and explain what you changed.

Review Image Generation Models (ai-3/10). Goal: Explain the **mechanism** and terminology of image generation and provenance.

Short practice: What may these two grids explain, and what can they not establish about a real generator?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

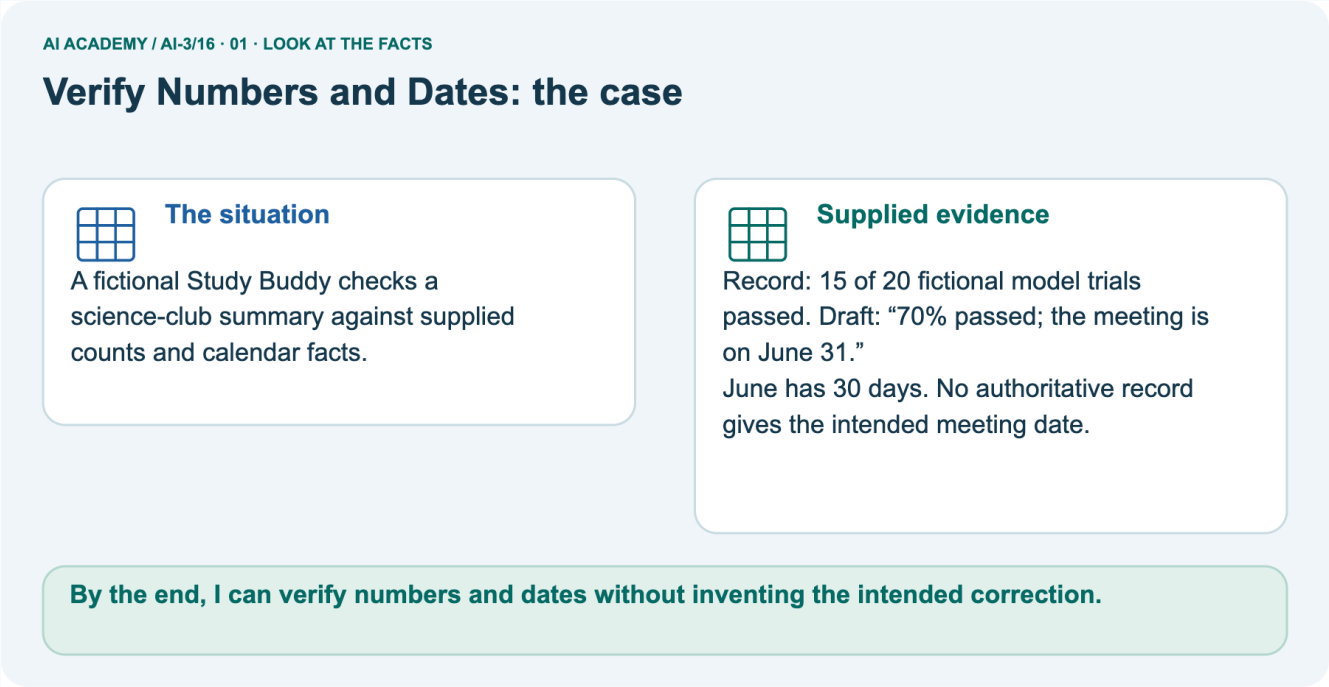
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Verify Numbers and Dates: the case



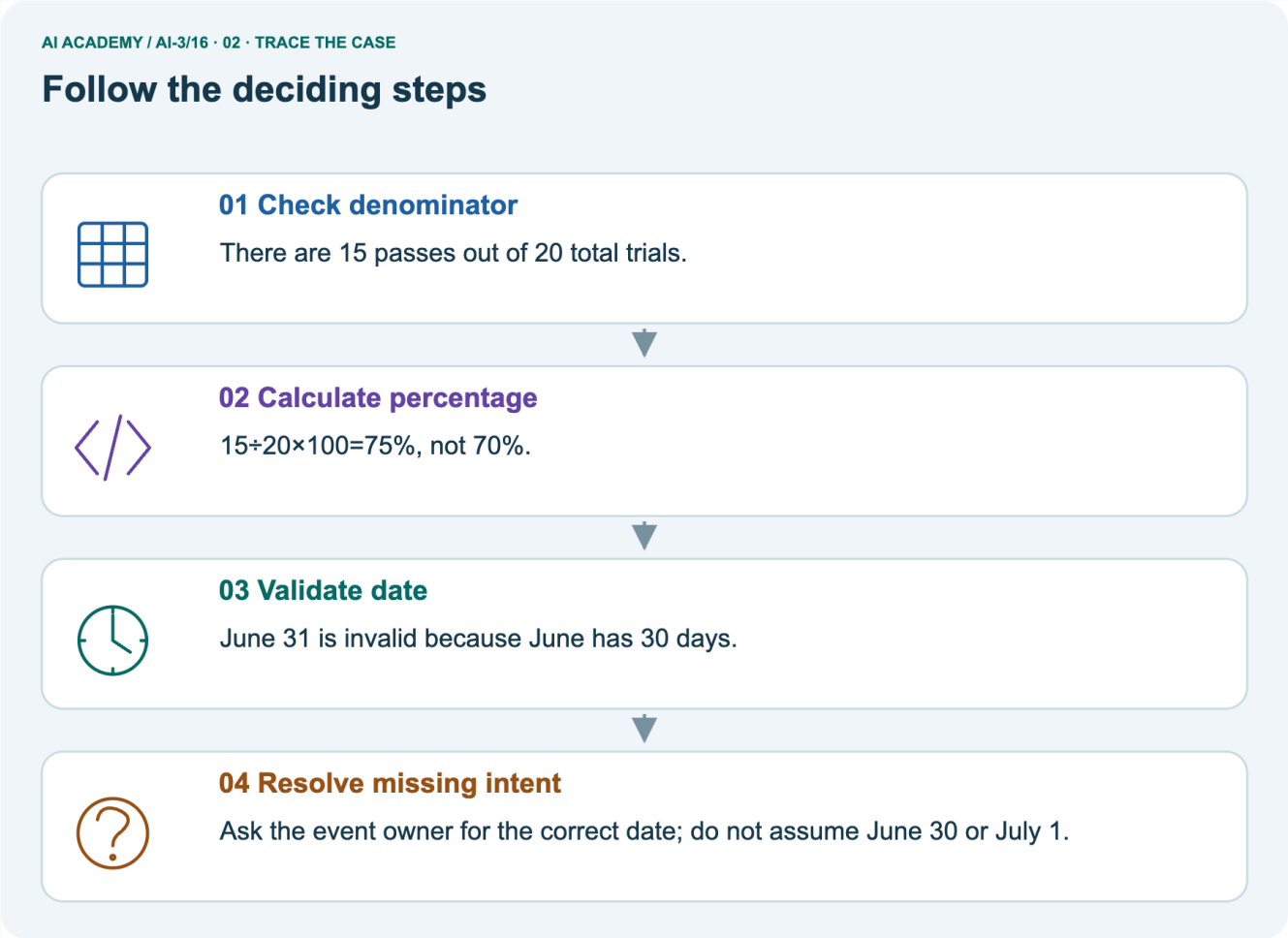
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts. Supplied facts: Record: 15 of 20 fictional model trials passed.

Draft: “**70%** passed; the meeting is on June 31.” June has 30 days. No authoritative record gives the intended meeting date.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

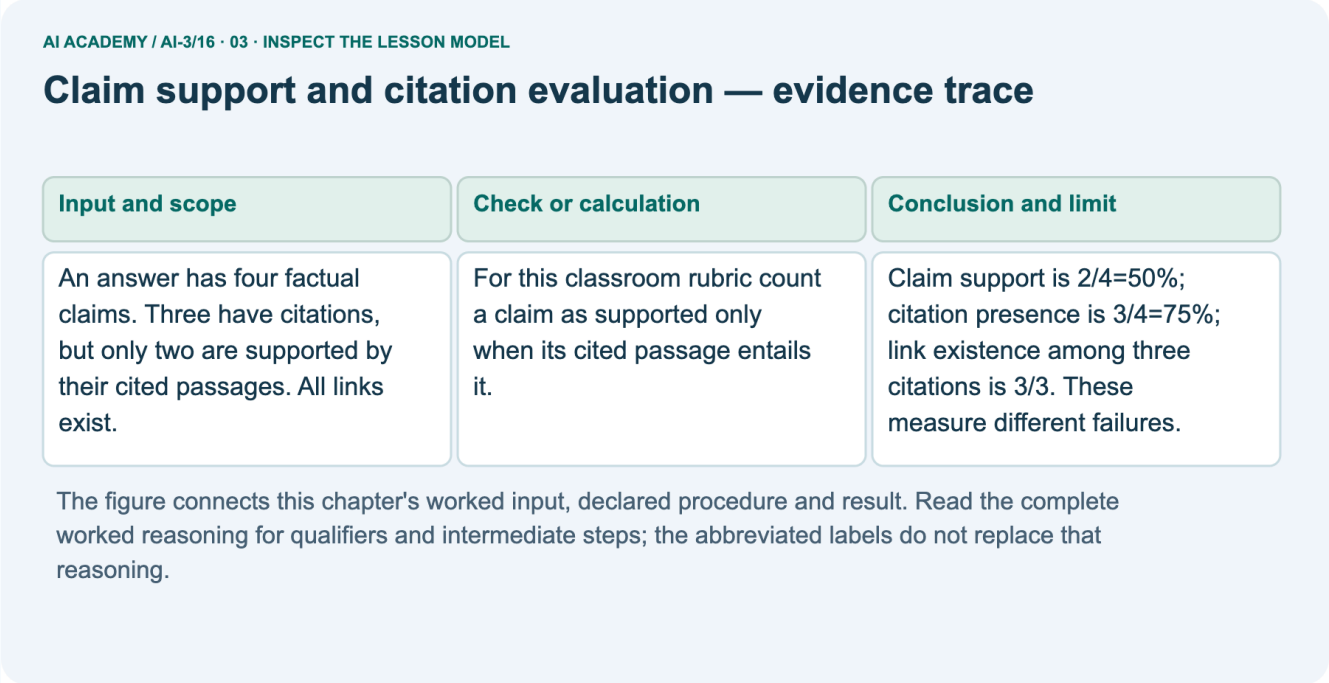


Follow the labelled picture with the matching explanation. Open the vector diagram

Check denominator: There are 15 passes out of 20 total trials. → Calculate percentage: 15÷20×100=75%, not **70%**. → Validate date: June 31 is invalid because June has 30 days. → Resolve missing intent: Ask the event owner for the correct date; do not assume June 30 or July 1.

Point to each numbered stage and explain which recorded **input** it uses.

03. Claim support and citation evaluation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

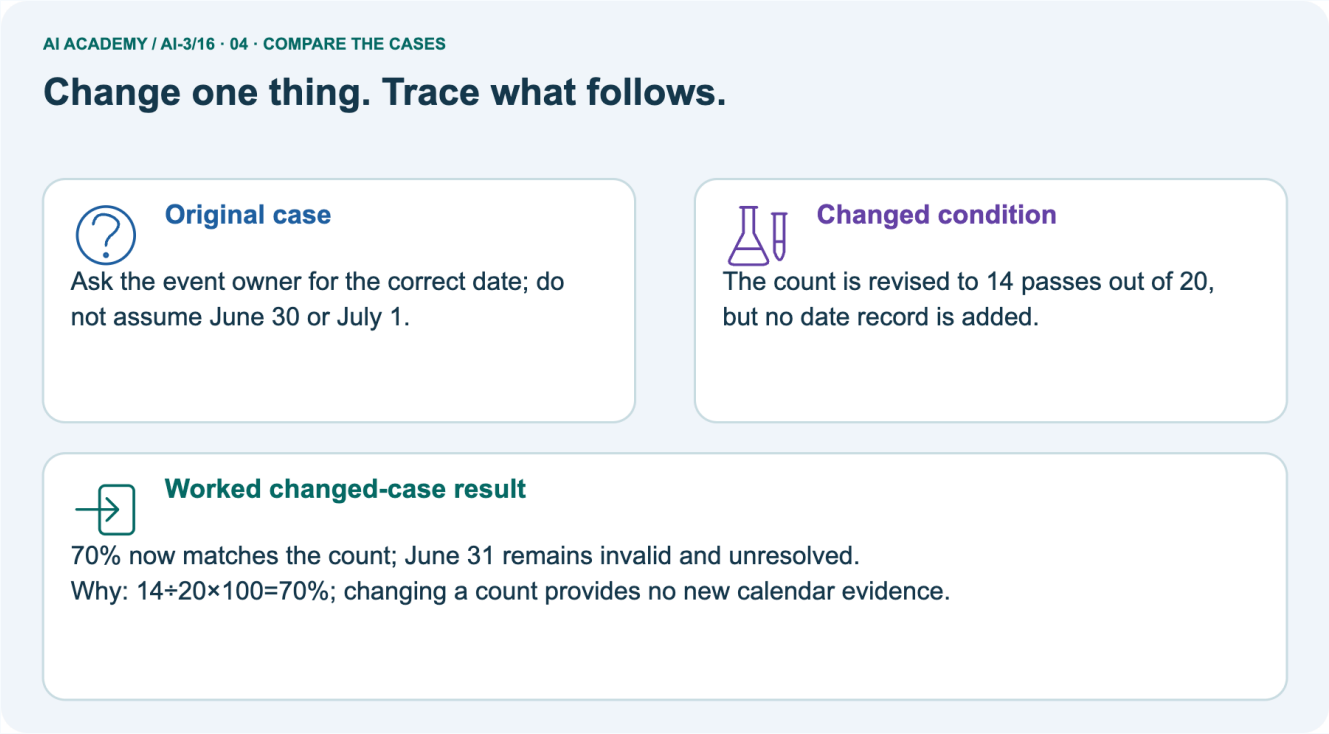
Claim support and citation evaluation — **evidence** trace. An answer has four factual claims. Three have citations, but only two are supported by their cited passages.

All links exist. | For this classroom rubric count a claim as supported only when its cited passage entails it. | Claim support is 2/4=50%; citation presence is 3/4=75%; link existence among three citations is 3/3.

These measure different failures.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



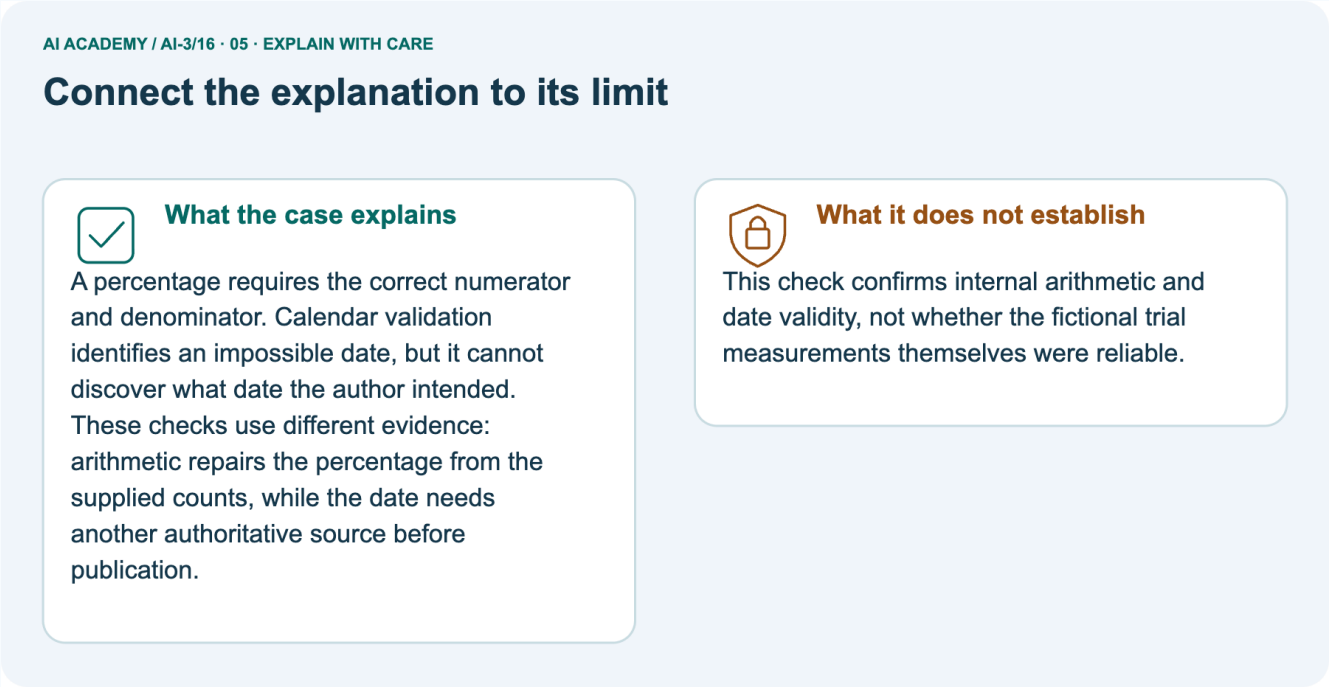
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Ask the event owner for the correct date; do not assume June 30 or July 1.
Changed condition: The count is revised to 14 passes out of 20, but no date record is added.

Expected result: **70%** now matches the count; June 31 remains invalid and unresolved.
Reason: 14÷20×100=**70%**; changing a count provides no new calendar **evidence**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

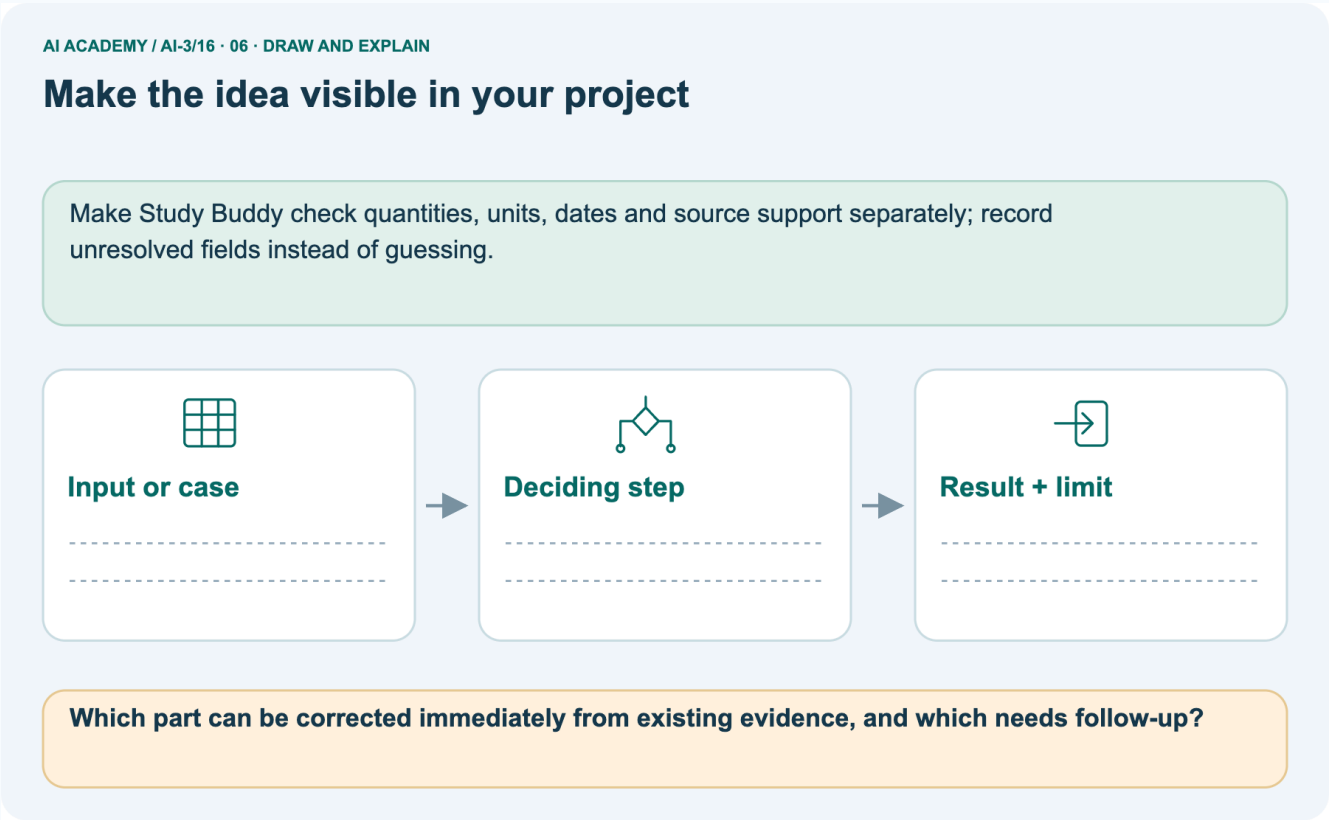
A percentage requires the correct numerator and denominator. Calendar validation identifies an impossible date, but it cannot discover what date the author intended.

These checks use different **evidence**: arithmetic repairs the percentage from the supplied counts, while the date needs another authoritative source before publication.

Limit: This check confirms internal arithmetic and date validity, not whether the fictional trial measurements themselves were reliable.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Make Study Buddy check quantities, **unit**s, dates and source support separately; record unresolved fields instead of guessing. Draw your **input**, deciding step, result and limit.

Supported question: Which part can be corrected immediately from existing **evidence**, and which needs follow-up?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Verify Numbers and Dates: the case

02. Follow the deciding steps

03. Claim support and citation evaluation — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.